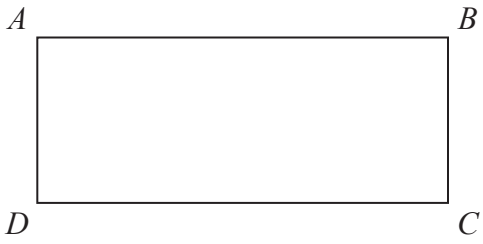
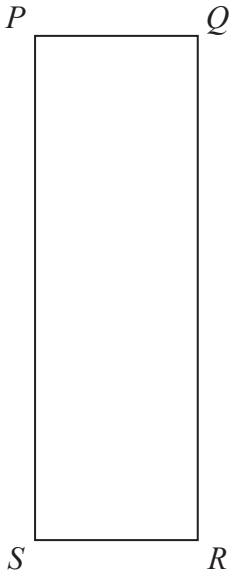
28 Here are two rectangles.





$QR = 10\,\text{cm}$
$BC = PQ$

The perimeter of $ABCD$ is 26 cm
The area of $PQRS$ is $45\,\text{cm}^2$

Find the length of AB.

PQ = 45 ÷ 10 = 4.5 cm, so BC = 4.5 cm.

1 mark
Use area ÷ height.

Let AB = x cm. Perimeter = 2x + 2 × 4.5

$$2x + 9 = 26$$
$$-9 \qquad -9$$
$$2x = 17$$
$$\div 2 \qquad \div 2$$
$$x = 8.5$$

1 mark
Use the perimeter 26 cm.

1 mark
Subtract 9, then divide both sides by 2.

1 mark
AB = 8.5 cm.

8.5 ..... cm

(Total for Question 28 is 4 marks)